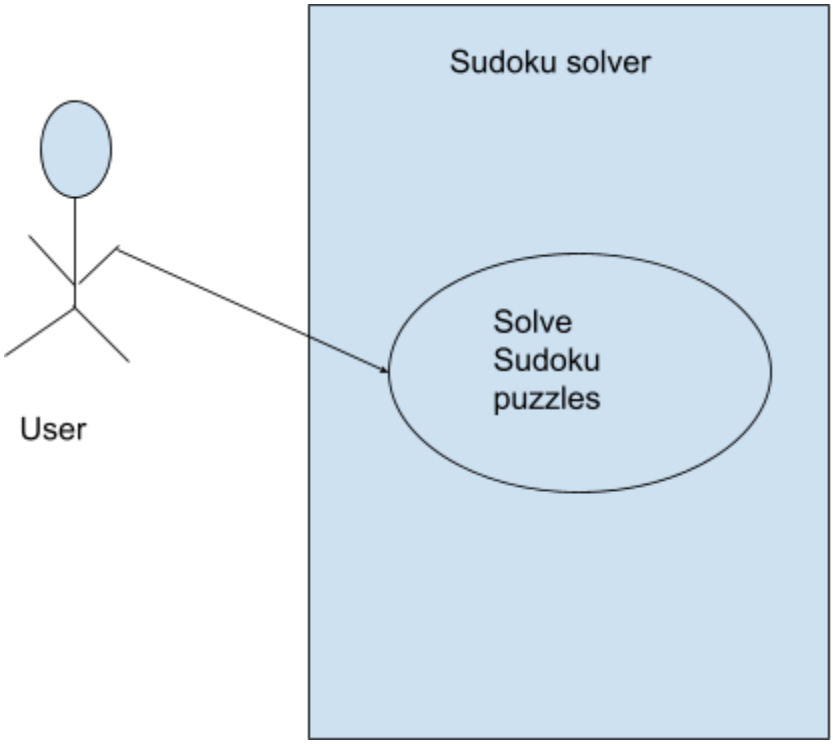

a. Use case - sudoku solver

b. Brief description = The User runs the program. It reads five puzzle files, solves each with recursive backtracking, and prints the file name, the puzzle, and every solution, or “No solution found.”

c. Flow
   1.The User runs the program.
   2.The system reads the next puzzle file.
   3.The system prints the file name and the puzzle.
   4.The system checks the starting digits for repeats.
   5.The system searches for solutions with backtracking.
   6.The system prints all solutions or “No solution found.”
   7.Repeat for all five files, then the program ends.

d. Precondition: puzzle1.txt to puzzle5.txt are in the program’s folder, with _ for blanks and numbers 1-9.
   Main flow: For each file, the system loads the 81 cells and prints the puzzle. It checks for repeated digits, then fills each empty cell with a valid digit from 1 to 9. If a cell has no valid digit, it undoes the last digit and tries the next one. It keeps searching after each solution, so it finds all of them. It then prints the solutions and the total.
   Exceptions: If a file is bad, or the puzzle repeats a digit, the system prints an error and skips that puzzle.
   Postcondition: The screen shows the results for all five puzzles.